



DPUSGETextureFFMpeg

Creates a dynamic video texture in the virtual world. The texture contains a video image. It is possible to display live video from a DirectShow / VideoForLinux compatible camera, such as a webcam. Additionally, a video file from the hard disk can be played.

1. Dynamic parameters:

<i>Play</i>	Only for video files: Starts video playback when the value changes from 0 to 1. Default: 1 (starts video playback immediately when the simulation is started).
<i>Stop</i>	Only for video files: Stops video playback when the value changes from 0 to 1. If the Play parameter is set again later, then the video file will restart from the beginning. Default: 0 (stops video playback only when the simulation is stopped)

2. Static parameters:

<i>View</i>	Name of the SGE view for which the video texture is updated. Additionally, it will be visible in all views that share textures with the specified SGE view. Default: <i>SGEView</i>
<i>TextureName</i>	Name of the created video texture, which must start with a dollar sign (\$). Default: <i>\$Video</i>
<i>Type</i>	Selects the type of video source to use. Default: (<i>empty value</i>) = use video file Alternative: <i>device</i> = use video camera
<i>Name</i>	Specifies the video file or device name. If a video file is used, this file name is searched within the SILAB installation directory and the project directory. If a video camera (device) is used, this is the name of the camera. The names of all connected video cameras are displayed if <i>Name</i> is empty.
<i>VideoW, VideoH</i>	Desired width and height of the video captured from the camera. If not specified, a camera-dependent default value is used. Warning: Most cameras accept only a number of fixed video formats. A similar format will be selected if the desired format is not available. Additionally, some formats may not work correctly although they are advised by the camera's driver.
<i>VideoFPS</i>	Desired frame rate of the video captured from the camera. If not specified, a camera-dependent default value is used.



Warning: Most cameras accept only a number frame rate. A similar frame rate will be selected if the desired one is not available. Additionally, some frame rates may not work correctly although they are advised by the camera's driver.

3. Functional principle:

Using *DPUSGETextureFFMpeg*, a dynamic video texture is created in the virtual world. The texture can contain a live video image that is captured from a DirectShow (on Windows) or VideoForLinux (on Linux) compatible camera. (Typical USB webcams should work.)

Alternatively, an existing video file from the hard disk can be played. Most of the commonly used video file formats can be played back. A list of supported formats can be found at <http://ffmpeg.org/general.html#Video-Codecs>

The *TextureName* parameter defines a name under which the video texture will be available in the SGE world. The name must start with a dollar sign (\$). The video texture can be used in any SGE object loaded after this DPU by using this texture name instead of specifying a texture file on the hard disk. Alternatively, *DPUSGEVideoObject* can be used to create a simple overlay object which displays the video texture.

4. Example of a SGE object which references a video texture:

```
define SIG3DObject VideoObj
{
    AlphaTest = 0.2;
    Blend = false;
    // [...]
    Texture = $MyVideo;
    TextureFilter = NoCompression | ClampS | ClampT | MagLinear |
                  MinLinear | Tex-ture;
    // [...]
};
```

5. Dependencies to other DPUs:

Please make sure that the computer on which **DPUSGETextureFFMpeg** runs also executes **DPUSGEWorld**, **DPUSGEObjectXXX** and **DPUSGEView**. The Index values of these DPUs must be set in the following order:

DPUSGEWorld.Index < DPUSGETextureFFMpeg.Index < DPUSGEObjectXXX.Index < DPUSGEView.Index

(DPUSGEObjectXXX stands for any DPU that uses the dynamic video texture in one of its displayed objects.)

6. Example:

The following script can be used to display a live video image from a video camera:



```
DPUSGETextureFFMpeg VidTexture
{
    Computer = {VIS};
    Index = 50;

    TextureName = "$MyVideo";
    Type = "device";
    # Example for Windows-based systems:
    # The camera name must be explicitly specified on Windows.
    # A list of possible
    # names will be displayed if Name is left empty.
    Name = "Integrated Webcam";
    # On Linux-based systems, Name can be left empty and
    # the first connected
    # camera will automatically be chosen. A list of possible
    # names will be
    # displayed if a Name is specified that was not found.
};
```

The following script can be used to display a video image from a video file:

```
DPUSGETextureFFMpeg VidTexture
{
    Computer = {VIS};
    Index = 50;

    TextureName = "$MyVideo";
    # The file is automatically searched within the
    # current project's directory.
    Name = "Videos/MyFirstVideo.mp4";
};
```